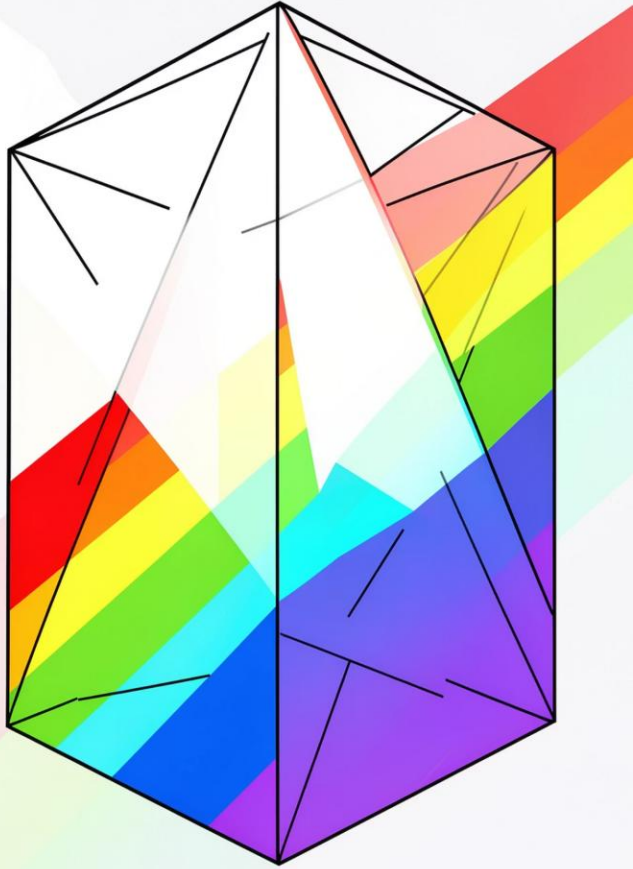




Physics Portfolio: Light & Electricity

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Subject Teacher: Sandhyarani Mam

Resistors in Parallel & Heating Effect

Resistors in Parallel

In a parallel circuit, resistors are connected across the same two points, providing multiple paths for current.

Key Characteristics

- Equivalent resistance: — reciprocals add.
- Equivalent resistance is always less than the smallest individual resistance.
- Current divides among branches:
- Voltage is the same across all resistors:
- If one resistor fails, the others continue to work.

Joule's Law of Heating

Joule's Law explains the heat produced when current flows through a resistor.

$$H = I^2Rt$$

where H is heat (Joules), I is current (Amperes), R is resistance (Ohms), and t is time (seconds).

Practical Applications



Electric Iron

A Nichrome heating element converts electrical energy into heat for ironing clothes.



Electric Toaster

Glowing resistance wires produce heat to toast bread evenly.



Electric Fuse

A thin wire melts when current exceeds the safe limit, protecting the circuit from damage.

Electric Power & Conclusion

Electric power is the rate at which electrical energy is transferred or consumed.

Key Formulas

- $P = VI$ — power = voltage $\times$ current
- $P = I^2R$ — power = current² $\times$ resistance, derived from Ohm's Law
- $P = V^2/R$ — power = voltage² $\div$ resistance, also derived from Ohm's Law

Meaning of Symbols

- P = power in Watts (W)
- V = voltage in Volts (V)
- I = current in Amperes (A)
- R = resistance in Ohms (Ω)

Units of Power and Energy

- SI unit of power: Watt (W)
- $1\text{ W} = 1\text{ J/s}$
- Commercial unit of energy: Kilowatt-hour (kWh), also called "Units"
- 1 kWh is the energy consumed by a 1000-Watt device operating for 1 hour
- $\text{Energy (kWh)} = \text{Power (kW)} \times \text{Time (hours)}$
- This is the unit used by electricity companies for billing.



Commercial Unit of Energy

The commercial unit of energy is the Kilowatt-hour (kWh), also called Units.
It is used by electricity companies to measure energy consumption and calculate electricity bills.

Calculation: $\text{Energy (kWh)} = \text{Power (kW)} \times \text{Time (hours)}$

Thank you, Sandhyarani Mam, for your excellent guidance in making the complex concepts of electricity and optics so understandable and interesting. Your teaching has made physics truly come alive.

- Viraj Chavan, Class 10B

Resistors in Series

In a series circuit, resistors are connected end-to-end in a single path, so current flows through each resistor in sequence.

Key Characteristics

- Total resistance: the sum of all individual resistances
- Current: constant throughout the circuit
- Voltage: divides among the resistors
- Failure mode: if one resistor opens, the entire circuit breaks

Core Formulas

$$R_S = R_1 + R_2 + R_3 + \dots$$

For a series circuit, total resistance is the sum of all resistors.

$$I_{total} = I_1 = I_2 = I_3$$

The same current flows through every component in the series path.

$$V_{total} = V_1 + V_2 + V_3$$

Voltage is shared across resistors according to their resistance values.

$$V = IR$$

Using Ohm's Law, the voltage across each resistor can be calculated from its current and resistance.

Advantages

- Simple to construct and analyze
- Easy to calculate total resistance
- Uses fewer branching connections

Disadvantages

- If one component fails, the whole circuit stops
- Voltage is divided, so devices may receive less than the source voltage

Applications

- Christmas lights in older strings
- Some alarm systems
- Basic circuit training and lab experiments

Resistivity

Resistivity (ρ , rho) is the specific electrical resistance of a material. It is an intrinsic property that depends only on the material itself and its temperature, not on its shape or size.

Definition

Resistivity is the electrical resistance of a unit cube of the material ($1\text{ m} \times 1\text{ m} \times 1\text{ m}$).

Formula

$$R = \rho \left(\frac{L}{A} \right)$$

where R is resistance, ρ is resistivity, L is length, and A is cross-sectional area.

SI Unit

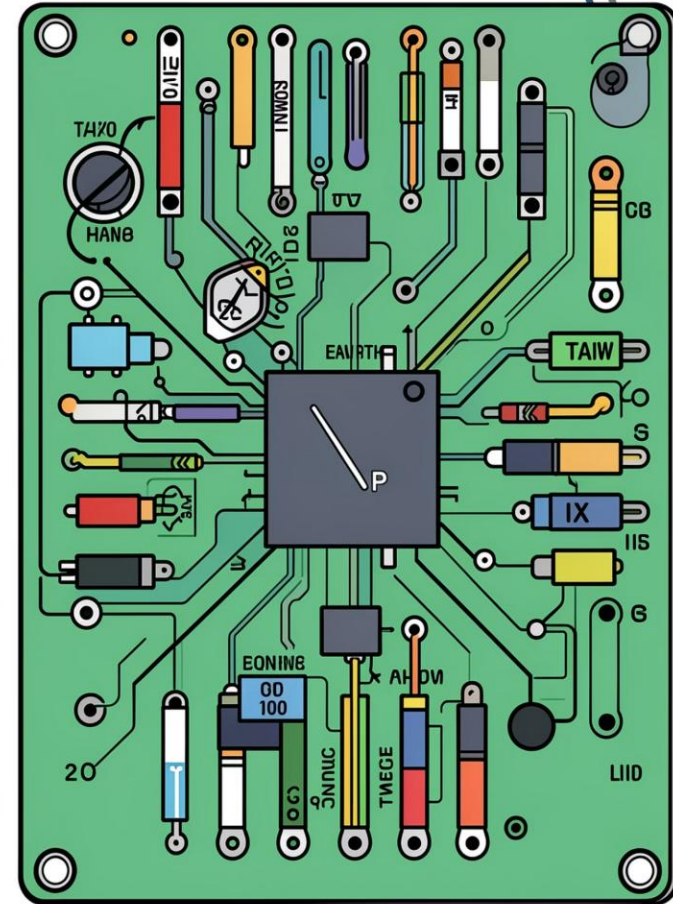
Ohm-meter ($l\Omega\text{megalcdot m}$).

Typical Values

- Copper $\approx 1.7 \times 10^{-8} \Omega \cdot \text{m}$ — good conductor
- Nichrome $\approx 1.0 \times 10^{-6} \Omega \cdot \text{m}$ — high resistivity

Practical Applications

Alloys such as Nichrome are widely used in electric heaters, toasters, and irons because their high resistivity produces significant heat when current passes through them, while their high melting point prevents them from melting.



Resistance & Factors Affecting It

Electrical resistance is the opposition offered by a material to the flow of electric current through it.

Factors affecting resistance:

01

Length (L)

Resistance is directly proportional to length, so a longer conductor has more resistance ($R \propto L$). Electrons face more obstacles along a longer path.

02

Cross-sectional area (A)

Resistance is inversely proportional to cross-sectional area, so a wider conductor has less resistance ($R \propto 1/A$). A larger area provides more paths for current flow.

03

Temperature

For most conductors, resistance increases as temperature rises. Higher temperature causes greater atomic vibrations, which hinder electron movement.

04

Nature of material (resistivity ρ)

Every material has an intrinsic property called resistivity. Different materials have different resistivities, so they offer different levels of opposition to current.

Combined formula:



Circuit Diagram Symbols

Standard electrical circuit symbols used in circuit diagrams:



Cell

Single long line and short line. Provides a single source of EMF.



Key / Switch

Line with a break or lever. Opens and closes the circuit.



Bulb

Circle with a cross inside. Converts electrical energy to light and heat.



Ammeter

Circle with the letter A inside. Measures current and is connected in series.



Battery

Multiple cells shown as parallel long and short lines. Provides higher voltage.



Wire Joint

Solid dot where wires connect. Indicates an electrical connection.



Resistor

Rectangle or zigzag line. Limits current flow.



Voltmeter

Circle with the letter V inside. Measures voltage and is connected in parallel.

Ohm's Law

Ohm's Law states that the electric current flowing through a conductor is directly proportional to the potential difference across it, provided the physical conditions, such as temperature, remain constant.

$$V = IR$$

Where: V is voltage (volts), I is current (amperes), and R is resistance (ohms).

Key relationships

- If voltage increases, current increases proportionally.
- If resistance increases, current decreases.

Conductors

- Ohmic conductors: materials that obey Ohm's Law, such as most metals at constant temperature.
- Non-ohmic conductors: materials that do not follow Ohm's Law, such as semiconductors and diodes.

The V-I graph for an ohmic conductor is a straight line passing through the origin, showing a linear relationship between voltage and current. The slope of the graph equals the resistance, R.

Electric Current

Electric current is the rate of flow of electric charge through a conductor.

Definition

Current is the amount of charge flowing past a point per unit time.

Formula

$$I = \frac{Q}{t}$$

I is current, Q is charge, and t is time.

SI Unit

Ampere (A), where $1 \text{ A} = 1 \text{ C/s}$.

Direction

By convention, current flows from the positive terminal to the negative terminal, opposite to electron flow.

Ammeter

Device used to measure electric current. It must be connected in series so all current passes through it.

Typical range: milliamperes (mA) to amperes (A).



Introduction to Electricity

Electric charge is a fundamental property of matter that causes it to experience a force in an electromagnetic field.

Properties of Charge

- Exists in two forms: positive charge (protons) and negative charge (electrons).
- Like charges repel each other; unlike charges attract.
- Charge is conserved — it cannot be created or destroyed, only transferred.
- Charge is quantized — it exists in discrete multiples of the elementary charge.

SI unit: Coulomb (C).

Quantization of charge: , where n is the number of electrons or protons and $e = 1.6 \times 10^{-19}$ C (elementary charge).

Reference: 1 Coulomb is approximately the charge of 6.24×10^{18} electrons.

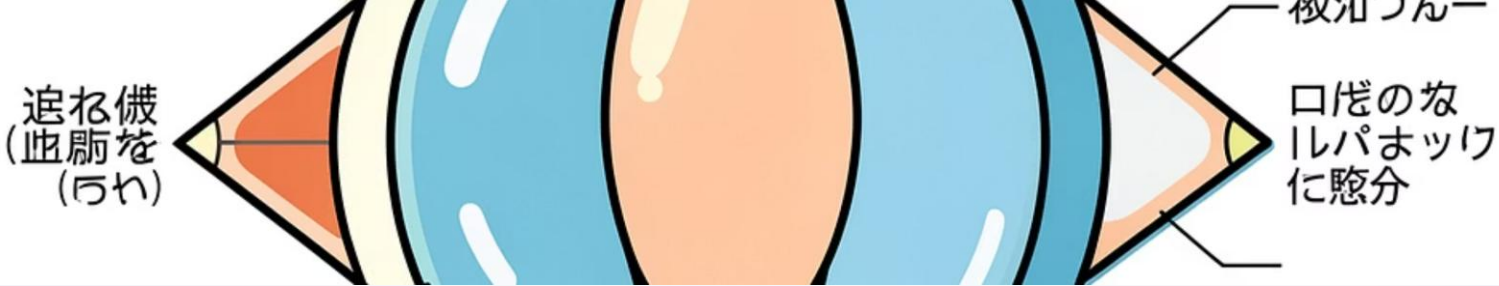


Image Formation by Concave Lenses

Concave lenses always form virtual, erect, and diminished images, regardless of object position.

Case 1: Object at Infinity

The image forms at the principal focus F_1 on the same side as the object. It is virtual, erect, and highly diminished, appearing point-sized.

Case 2: Object at Finite Distance

The image forms between F_1 and the optical center O , on the same side as the object. It remains virtual, erect, and diminished.

📌 Key point: The image formed by a concave lens is always on the same side as the object.

Practical Applications

- Corrective lenses for myopia: Concave lenses diverge incoming light rays so they focus correctly on the retina.
- Door peepholes: Provide a wide field of view for safe viewing.
- Binoculars and telescopes: Used in combination with other lenses to adjust and refine the light path.

Electric Potential & Potential Difference

Electric potential at a point is defined as the work done per unit positive test charge to move it from infinity to that point against the electric field.

$$V = \frac{W}{Q}$$

Where V is potential, W is work done, and Q is charge.

SI unit: volt (V), named after Alessandro Volta.

1 volt = 1 joule per coulomb ($1V = 1J/C$).

Potential difference (voltage) between two points is the difference in electric potential:

$$V_{AB} = V_A - V_B$$

It tells us how much energy is transferred per unit charge when charge moves between two points in a circuit.

Voltmeter

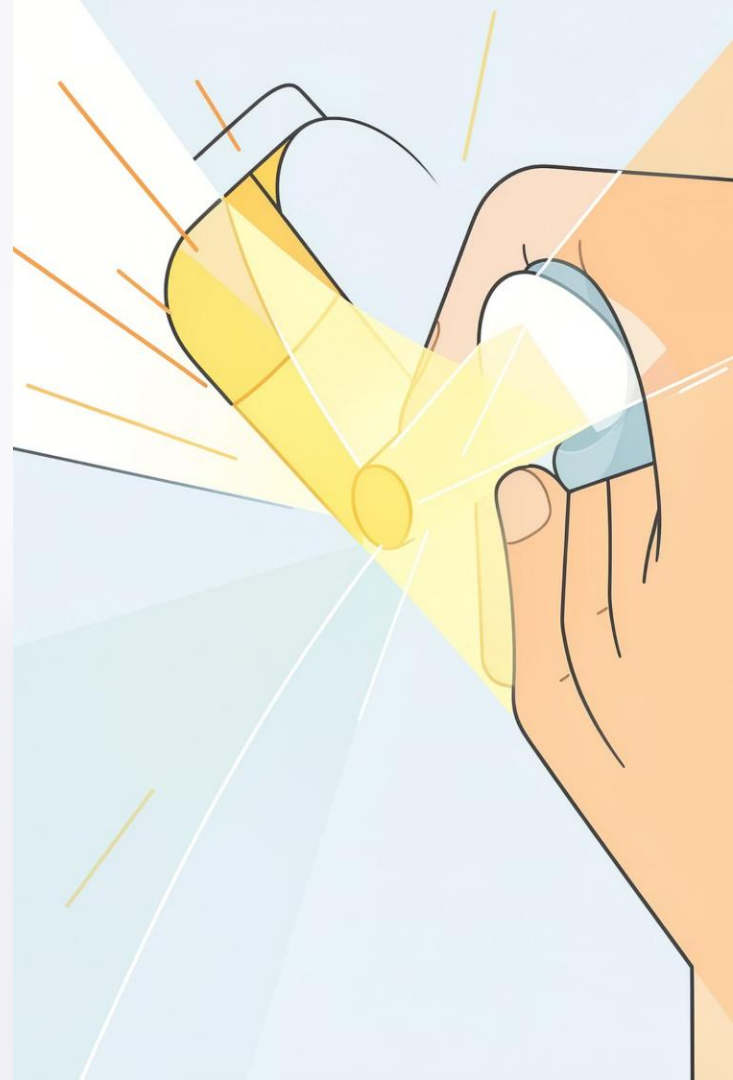
- **Purpose:** Measures potential difference between two points.
- **Connection:** Must be connected in parallel across the component, not in series.
- **Typical range:** From millivolts (mV) to volts (V).

Image Formation by Convex Lenses

Convex lenses can form both real and virtual images depending on the object position.

Object Position	Image Position	Nature	Orientation	Size
Object at infinity	At F_2	Real	Inverted	Point-sized
Beyond $2F_1$	Between F_2 and $2F_2$	Real	Inverted	Diminished
At $2F_1$	At $2F_2$	Real	Inverted	Same size
Between F_1 and $2F_1$	Beyond $2F_2$	Real	Inverted	Enlarged
At F_1	At infinity	Real	Inverted	Infinitely large
Between F_1 and O	On same side as object	Virtual	Erect	Enlarged

i Real images form when the object is beyond F_1 ; virtual images form only when the object is between F_1 and the lens.



Lens Formula & Power of a Lens

Lens formula relates object distance (u), image distance (v), and focal length (f):

$$1/f = 1/v - 1/u$$

Note: This is different from the mirror formula.

Magnification is given by:

$$m = v/u$$

Note: Magnification is positive for virtual images and negative for real images.

Power of a lens is the reciprocal of its focal length:

$$P = 1/f$$

where f is measured in meters.

The SI unit of power is the Dioptre (D).

Practical examples:

- A lens with $f = +0.5 \text{ m}$ has power $P = +2 \text{ D}$ and is converging.
- A lens with $f = -0.5 \text{ m}$ has power $P = -2 \text{ D}$ and is diverging.

❗ Dioptre is commonly used to specify lens strength in spectacles and optical instruments.





Rules for Ray Diagrams (Lenses)

Three principal rays are used to construct ray diagrams for lenses. By tracing any two of these rays, the image position can be located where the rays intersect, or where their extensions appear to intersect.

01

Ray Parallel to the Principal Axis

For a convex lens, a ray parallel to the principal axis refracts through the focal point F_2 on the opposite side of the lens. For a concave lens, the refracted ray diverges and appears to originate from the focal point F_1 on the same side as the incident ray.

02

Ray Through the Focal Point

For a convex lens, a ray passing through the focal point F_1 before refraction emerges parallel to the principal axis. For a concave lens, a ray directed toward the focal point F_2 emerges parallel to the principal axis after refraction.

03

Ray Through the Optical Center

For both convex and concave lenses, a ray passing through the optical center O travels undeviated, continuing in a straight line.

Spherical Lenses: Introduction

Spherical lenses are transparent optical devices that refract light to form images. Although both types follow the same laws of refraction, their shapes determine how they bend light and the kind of image they produce.

Convex lens (converging lens)

- Thicker at the center and thinner at the edges
- Converges parallel rays to a single focal point
- Used in magnifying glasses, cameras, projectors, and corrective lenses for hyperopia (farsightedness)

Concave lens (diverging lens)

- Thinner at the center and thicker at the edges
- Diverges parallel rays, making them appear to come from a virtual focal point
- Used in corrective lenses for myopia (nearsightedness), door peepholes, and binoculars

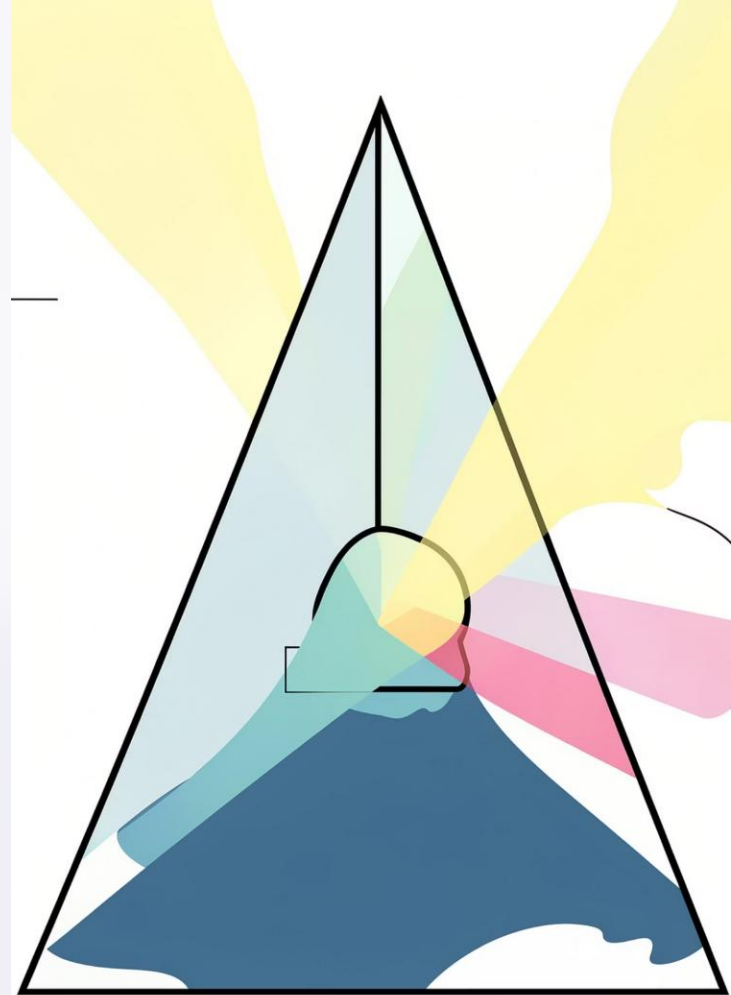
Laws of Refraction & Snell's Law

Two fundamental laws of refraction

- **First Law of Refraction:** The incident ray, refracted ray, and the normal to the interface all lie in the same plane.
- **Second Law of Refraction (Snell's Law):** For any two given media, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant and equals the relative refractive index of the second medium with respect to the first.

$$\frac{\sin(i)}{\sin(r)} = n_{21}$$

Snell's Law is crucial for understanding how lenses, prisms, and other optical instruments work.



Refractive Index

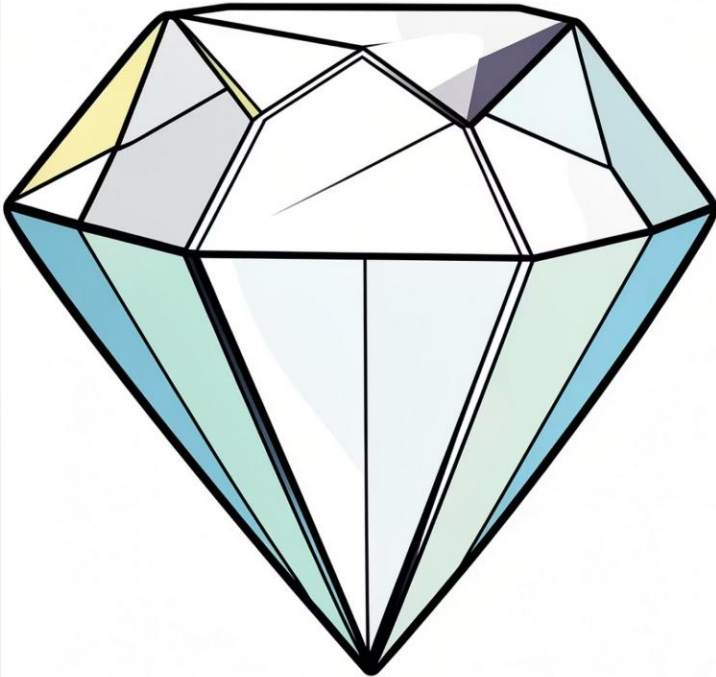
Absolute refractive index (n) of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in that medium:

where $c = 3 \times 10^8 \text{ m/s}$ and v is the speed of light in the medium. For any material medium, $n > 1$.
Relative refractive index (n_{21}) of medium 2 with respect to medium 1 is:

This tells us how much light bends when passing from one medium to another.

Refractive indices of common materials

Material	Refractive Index	Notes
Air	≈ 1.00	Very close to vacuum
Water	≈ 1.33	Light slows moderately
Glass	≈ 1.50	Common optical material
Diamond	≈ 2.42	Highest among common materials; responsible for exceptional brilliance and sparkle



Refraction through a Glass Slab

When light passes through a rectangular glass slab, it refracts at both surfaces. The emergent ray is parallel to the incident ray, but is shifted sideways.

Path of the ray

As the light enters the glass, it bends towards the normal. On leaving the slab, it bends away from the normal, producing a parallel emergent ray.

This happens because the slab has parallel faces, so the final direction matches the original direction even though the ray is displaced laterally.



Key angles and relationships

- Angle of incidence (i) — angle between the incident ray and the normal at the first surface.
- Angle of refraction (r) — angle between the refracted ray and the normal inside the glass. Since glass is optically denser than air, $r < i$.
- Angle of emergence (e) — angle between the emergent ray and the normal at the second surface. For a slab with parallel surfaces, $e = i$.
- Lateral displacement — the perpendicular distance between the incident ray produced forward and the emergent ray.

Important: The emergent ray is parallel to the incident ray, but it does not follow the same path. It is shifted sideways due to refraction at both surfaces.



Mirror Formula & Magnification

Mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

Magnification: $m = -v/u = h'/h$

Relation: $R = 2f$

Solved Example

Object 5cm tall at 30cm from a concave mirror with $f = 10\text{cm}$. Find the image position and size.

1. Use mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
2. Substitute values: $\frac{1}{(-10)} = \frac{1}{v} + \frac{1}{(-30)}$
3. Solve for v : $\frac{1}{v} = \frac{1}{(-10)} - \frac{1}{(-30)} = -\frac{1}{15}$, so $v = -15\text{cm}$
4. Find magnification: $m = -v/u = -(-15)/(-30) = -\frac{1}{2}$
5. Find image height: $h' = mh = (-\frac{1}{2})(5) = -2.5\text{cm}$

Final answer: The image is real, inverted, 15cm in front of the mirror, and 2.5cm tall.

Refraction of Light

Refraction is the bending of light when it passes from one transparent medium to another.

Optical density

A measure of how much a medium slows down light.

Speed of light in media

Light travels fastest in vacuum at 3×10^8 m/s, and slower in denser media such as water and glass.

Why light bends

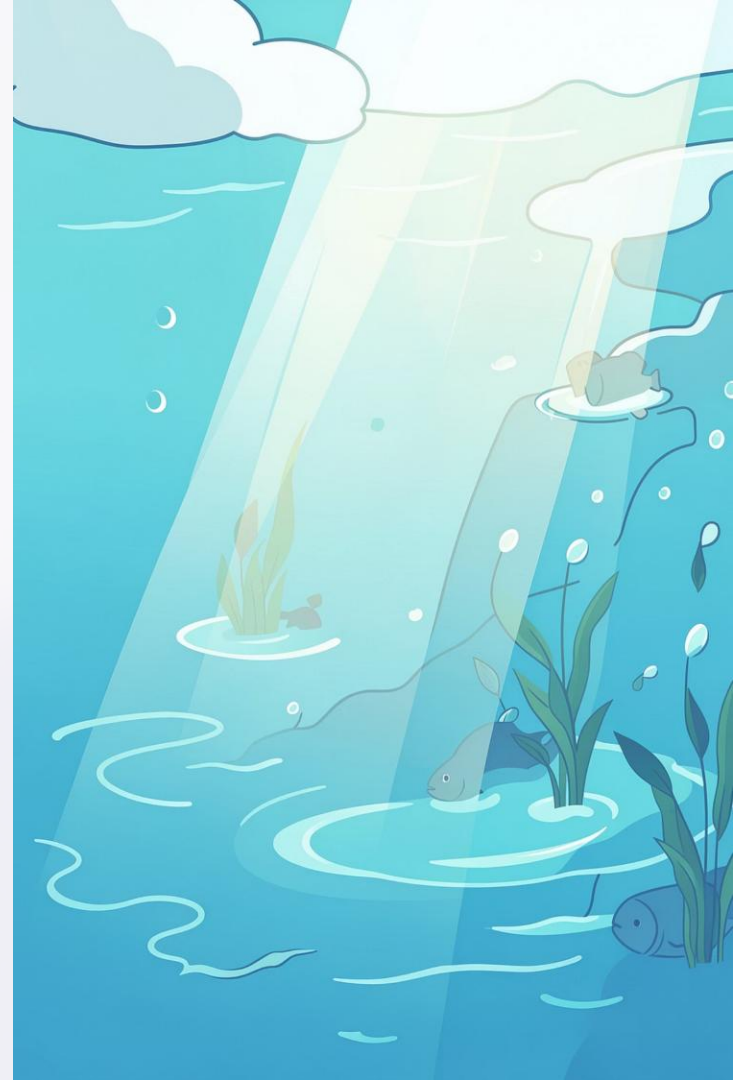
When light enters a denser medium at an angle, the part of the wavefront that enters first slows down before the rest, causing the ray to bend toward the normal.

Direction of bending

Light bends toward the normal when entering a denser medium and away from the normal when entering a less dense medium.

No bending at 90°

If light enters perpendicular to the surface, its speed changes but its direction does not.



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Sign Convention for Spherical Mirrors

Cartesian Sign Convention

Light travels from left to right.

All distances are measured from the pole (P).

Distances in the direction of incident light are positive.

Distances opposite to the direction of incident light are negative.

Sign Rules for Mirrors

Parameter	Concave Mirror	Convex Mirror
Object distance (u)	Negative	Negative
Image distance (v)	Negative for real image, positive for virtual image	Positive for virtual image
Focal length (f)	Negative	Positive
Radius of curvature (R)	Negative	Positive

Convex Mirror Ray Diagrams & Uses

Convex mirrors always form virtual, erect, diminished images.

Image Formation

Case 1: Object at infinity

Image forms at focus F, is virtual and erect, and is point-sized.

Case 2: Object at finite distance

Image forms between pole P and focus F, is virtual and erect, and is diminished.

Practical Applications

Concave Mirrors

- Shaving and makeup mirrors for a magnified erect image
- Headlights to produce a parallel beam
- Solar furnaces to concentrate heat

Convex Mirrors

- Rear-view mirrors in vehicles for a wide field of view
- Security mirrors in shops
- Road safety mirrors at bends and blind corners

Concave Mirror Ray Diagrams (Part 1)

Three important cases for concave mirrors:

Object position	Image position	Nature	Size
Object at infinity	At principal focus F	Real, inverted	Point-sized; highly diminished
Object beyond center of curvature C	Between F and C	Real, inverted	Diminished
Object at center of curvature C	At C	Real, inverted	Same size as object

Concave Mirror Ray Diagrams (Part 2)

Three more important cases for concave mirrors.

Case 4: Object between C and F

Object position: Between C and F

Image position: Beyond C

Nature: Real and inverted

Size: Enlarged

Case 5: Object at focus F

Object position: At F

Image position: At infinity

Nature: Real and inverted

Size: Infinitely large

Case 6: Object between P and F

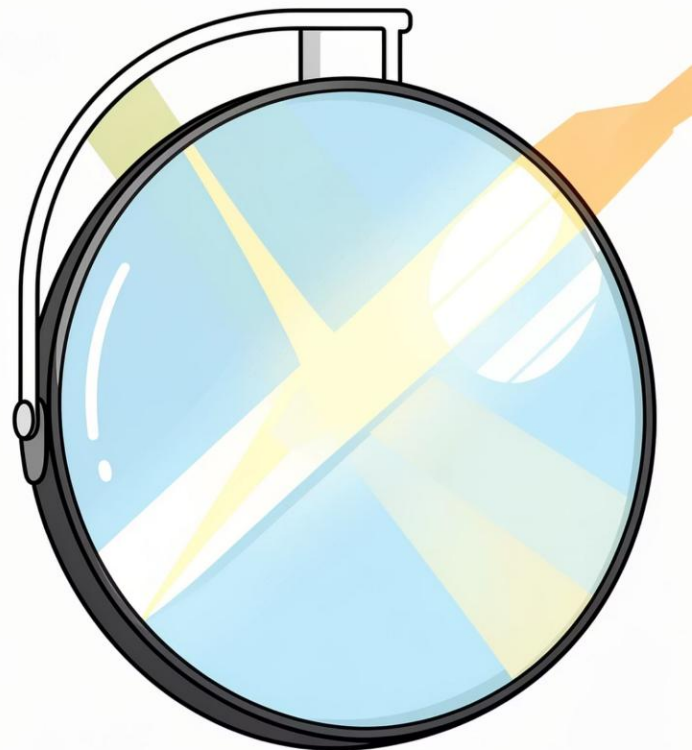
Object position: Between P and F

Image position: Behind the mirror

Nature: Virtual and erect

Size: Enlarged

The image changes from real to virtual as the object moves from beyond the focal point to between the pole and the focus.



Rules for Ray Diagrams (Mirrors)

Four standard rules for tracing rays reflected by spherical mirrors:

01

1. Parallel ray

A ray parallel to the principal axis reflects through the principal focus in a concave mirror, or appears to come from the focus in a convex mirror.

02

2. Focus ray

A ray passing through the principal focus reflects parallel to the principal axis.

03

3. Center ray

A ray passing through the center of curvature reflects back along the same path.

04

4. Pole ray

A ray striking the pole reflects at an angle equal to the angle of incidence.

Use these rules together to determine image position and nature.



Terminology of Spherical Mirrors

Key terms for spherical mirrors:

Pole (P)

The geometric center of the mirror's reflecting surface.

Center of Curvature (C)

The center of the sphere from which the mirror is a part.

Radius of Curvature (R)

The distance from the pole to the center of curvature.

Principal Axis

The straight line passing through the pole and the center of curvature.

Principal Focus (F)

The point where parallel rays converge for a concave mirror, or appear to diverge from for a convex mirror.



Important relation: $R = 2f$

The radius of curvature is twice the focal length.

Reflection of Light

Reflection is the phenomenon in which light bounces back from a surface after striking it.

The two Laws of Reflection:

- 1 The angle of incidence equals the angle of reflection ($\angle i = \angle r$).
- 2 The incident ray, reflected ray, and the normal all lie in the same plane.

Types of Reflection

Regular Reflection

Occurs on smooth surfaces like mirrors. The reflected rays remain parallel, producing a clear, sharp image.

Diffused Reflection

Occurs on rough surfaces like walls. The reflected rays scatter in many directions, so no clear image is formed.

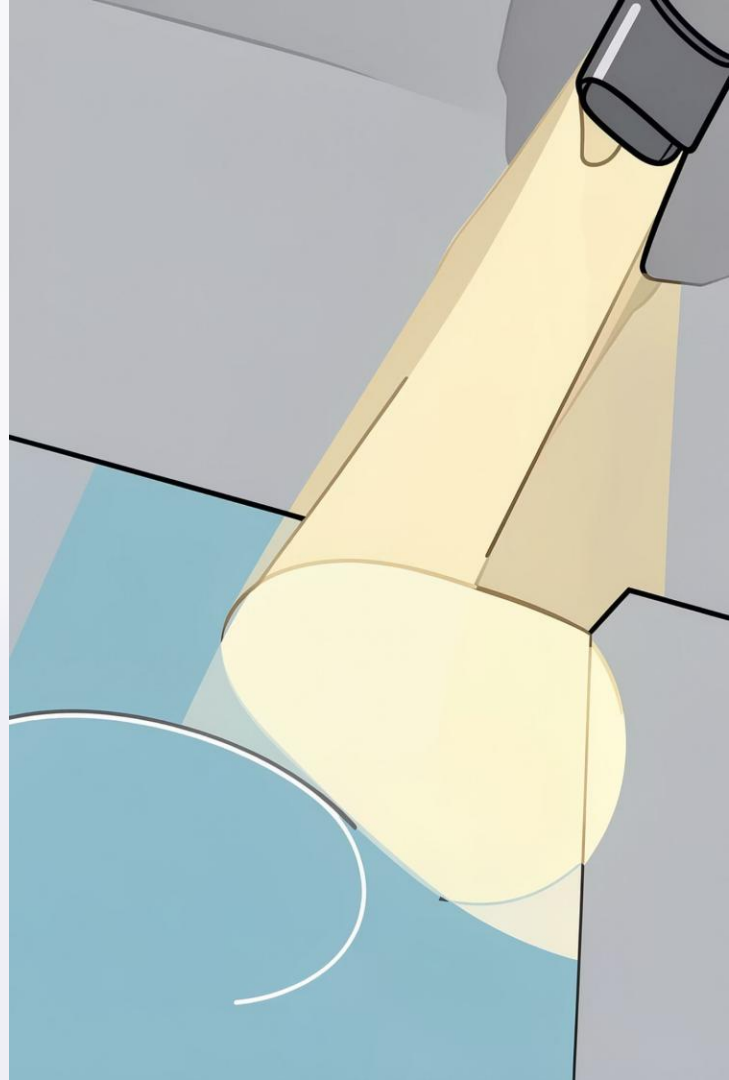
Quick Comparison

Regular Reflection

- Smooth surface
- Parallel reflected rays
- Clear image formation

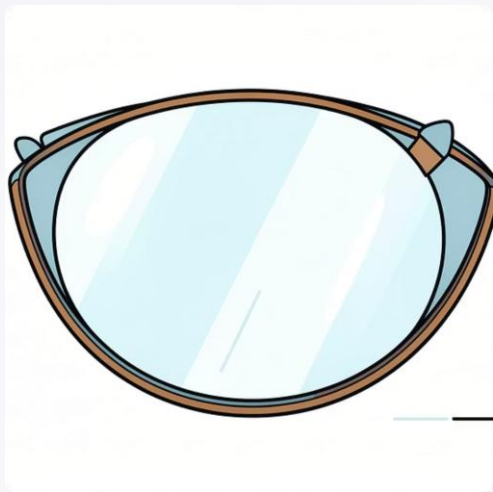
Diffused Reflection

- Rough surface
- Scattered reflected rays
- No clear image formation



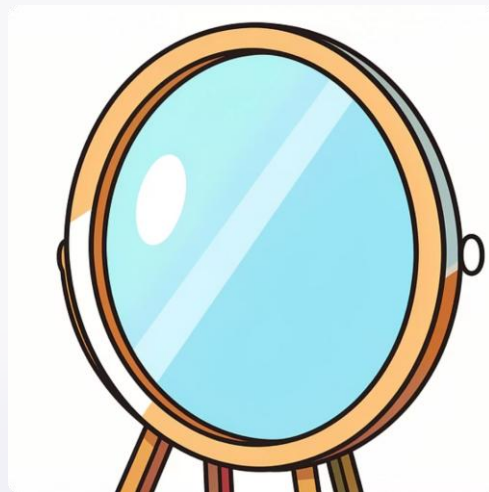
Spherical Mirrors Introduction

A spherical mirror is a mirror whose reflecting surface forms part of a hollow sphere.



Concave Mirror

Converging mirror; the reflecting surface curves inward like the inside of a bowl. It brings light rays to a focal point and is used in shaving mirrors and headlights.



Convex Mirror

Diverging mirror; the reflecting surface curves outward like the outside of a sphere. It spreads light rays outward and is used in rear-view mirrors for a wider field of view.

Introduction to Light

Light is a form of electromagnetic energy that enables vision.

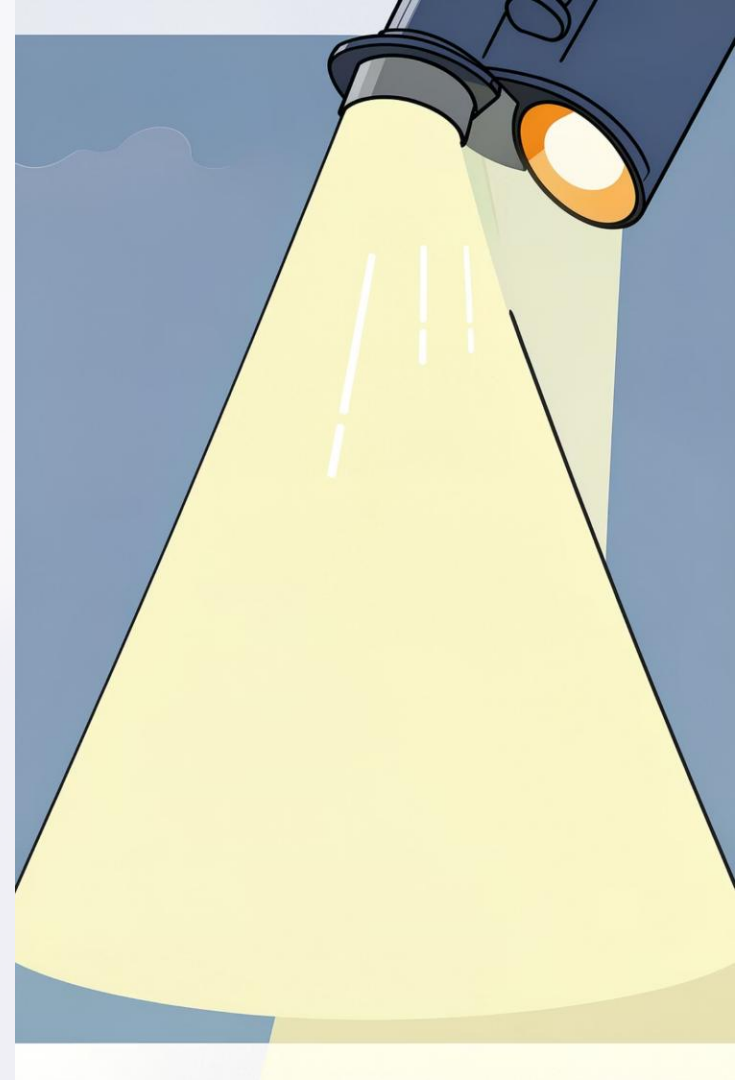
Properties of light

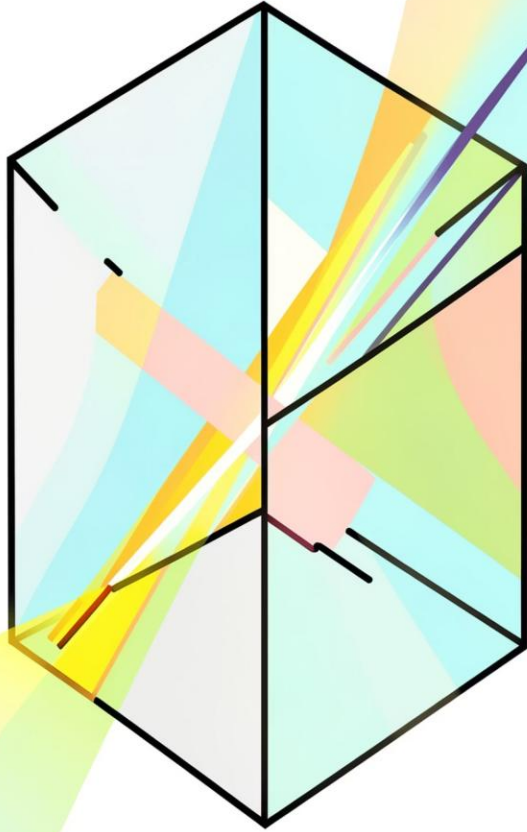
- Travels in straight lines (rectilinear propagation)
- Does not require a medium and travels through vacuum at 3×10^8 m/s
- Can be reflected, refracted, and diffracted

Dual nature of light

Light exhibits both wave and particle (photon) behavior.

- Wave nature: interference and diffraction
- Particle nature: photoelectric effect





Physics Portfolio: Light & Electricity

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